StockEvent Lens — Batch 019 Intelligence Dashboard

Articles 091–095 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_19 (Articles 091-095)`

TEST 091
Original — 100% Base
NVIDIA market cap reaches $5.4T on Sept 25, exceeding Russell 2000 total cap by $2T.

TEST 092
Repeat — 95% Overlap
Synthetic repeat detailing identical $5.4T valuation and $193B vs -$13B earnings split.

TEST 093
Update — +18% New Info
Advances state to Sept 28 valuation of $5.45T, lifting Vanguard S&P 500 ETF weight to 8.08%.

TEST 094
Conflict — Dispute ($6.2T)
Contradicts Sept 25 benchmark: claims NVIDIA reached $6.2T, surpassing Russell by $2.7T.

TEST 095
Different — 0% Overlap
AMD technical breakout above $550 to $642.49 with Nov $600/$750 call spread starting Group 038.

Technical Analysis: Dense Vector Blindness vs. Numerical Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *numerical or market valuation accuracy*. Changing valuation amounts (e.g., $5.4 trillion vs $6.2 trillion market cap on Sept 25 in test_094) yields high vector similarity (~0.92) because both texts share identical surrounding context tokens ("NVIDIA", "market capitalization", "Russell 2000", "September 25").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard date-specific numeric valuation dispute.
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "$5.4 trillion" and "$6.2 trillion" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this valuation metric conflict.

2. LLM Evaluation Override
To prevent missing critical valuation disputes, the LLM reasoning layer performs explicit key figure attribute-value matching, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected.

Article Relationship & Percentage Breakdown (Batch 019)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_091_original`	Original (%5=1)	NVIDIA $5.4T cap, Russell 2000 $3.5T, Charlie Bilello, $193B profit vs $13B loss	100%	Baseline	Establishes breaking story frame on NVIDIA's $5.4T valuation landmark vs Russell 2000 small-cap benchmark.
`test_092_repeat`	Repeat (%5=2)	NVIDIA $5.4T market value, Russell 2000 $3.5T, Bilello earnings contrast	95%	0% New Facts	Restates identical valuation metrics, benchmark gap, and earnings comparison without adding new details.
`test_093_update`	Update (%5=3)	**Sept 28 $5.45T valuation**, Vanguard S&P 500 ETF (VOO), 8.08% weight	82%	+18% Fund Metrics	Advances event state to Sept 28 market movement ($5.45T cap) and updates VOO fund weighting to 8.08%.
`test_094_conflict`	Conflict (%5=4)	**$6.2T valuation claim**, Sept 25 date dispute, $2.7T gap contest	92% Match	$\Delta F_{contradiction}$ = 1.0	Directly disputes Sept 25 baseline valuation, asserting NVIDIA reached $6.2T rather than $5.4T.
`test_095_different`	Different (%5=0)	AMD breakout $642.49, $1T cap milestone, OptionsPlay Tony Zhang Nov $600/$750 call spread	0%	Group 038 Trigger	Zero domain overlap with NVIDIA index comparison; triggers standalone Group 038 technical event cluster.

Group 037 — Consolidated Event Summary

4 Articles (091–094)

NVIDIA's market capitalization reached $5.4 trillion on Sept 25, surpassing the Russell 2000's $3.5 trillion value by nearly $2 trillion based on Charlie Bilello's analysis highlighting NVIDIA's $193B profit versus $13B in small-cap losses (Articles 091–092). Valuation state progressed as subsequent price movement pushed NVIDIA's valuation to **$5.45 trillion on Sept 28, driving its weighting in the Vanguard S&P 500 ETF (VOO) to 8.08%** (Article 093). However, market capitalization figures are in direct conflict: original reports confirmed Sept 25 valuation of $5.4 trillion ($1.9T above Russell 2000), whereas a contradictory report claims **NVIDIA reached $6.2 trillion on Sept 25, outvaluing the Russell 2000 by $2.7 trillion** (Article 094 ; $\Delta F_{contradiction}$ = 1.0).

✓ Update Preserved: NVIDIA valuation rose to $5.45T on Sept 28, elevating its VOO fund weight to 8.08%

✓ Conflict Preserved: Sept 25 valuation of $5.4T (091/092) versus $6.2T (094)

✓ Duplicate Removed: Article 092 Consolidated

Group 038 — Standalone Summary

1 Article (095)

Advanced Micro Devices (AMD) broke above its $550 consolidation ceiling to reach $642.49 in premarket trading after joining the $1 trillion market cap club, with OptionsPlay strategist Tony Zhang recommending a Nov $600/$750 call spread to target $750 (Article 095).

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 037

Batch 019 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

`test_091_original_group_037.json` — Group 037

```
{
  "id": 62016564,
  "headline": "Nvidia's $5.4 Trillion Market Cap 'Seem
  "symbols": ["AAPL", "AMZN", "AVGO", "GOOG", "GOOGL",
  "summary": "Charlie Bilello notes Nvidia's $5.4T marl
}
```

Establishes $5.4T market cap vs Russell $3.5T baseline.

`test_093_update_group_037.json` — Group 037

```
{
  "id": 990000093,
  "headline": "NVIDIA weight in Vanguard S&P 500 ETF r
  "symbols": ["NVDA", "VOO"],
  "summary": "NVIDIA's weight in broad market index fu
}
```

Adds Sept 28 $5.45T valuation & VOO 8.08% weight.

`test_094_conflict_group_037.json` — Group 037

```
{
  "id": 990000094,
  "headline": "NVIDIA market cap reaches $6.2 trillion
  "symbols": ["NVDA"],
  "summary": "NVIDIA achieved a market capitalization
}
```

Disputes Sept 25 valuation ($6.2T vs $5.4T).